

Fullstack Development

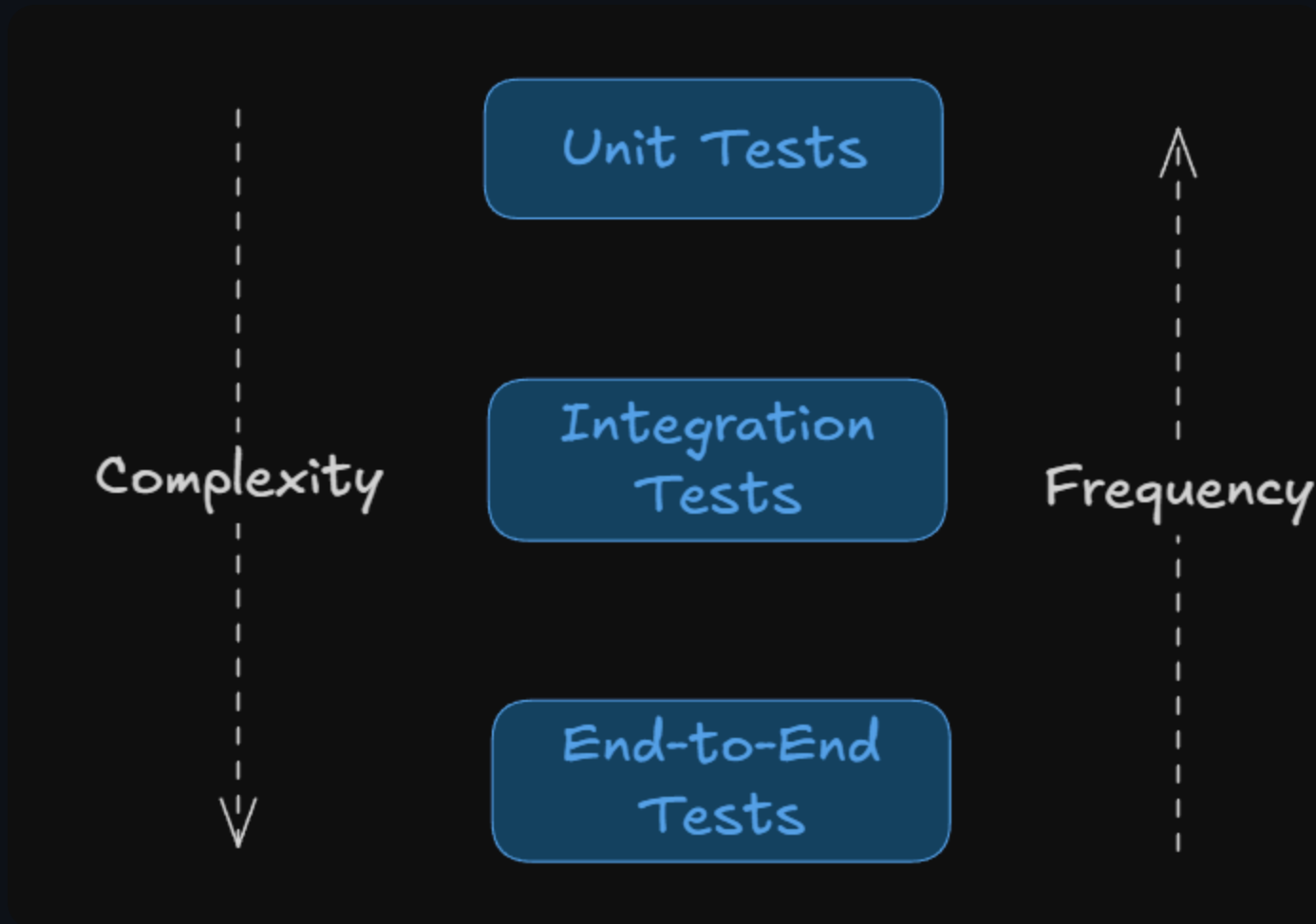
Preflight project - automated testing

- Vitest
- Cypress

Setup (Repos)

- DB
- Backend
- Frontend

Testing types



Testing types

- Unit test
 - Tests individual code components.
- Iteration test
 - Checks code dependency.
- End-to-end test
 - Assesses an application's functionality and user experience.

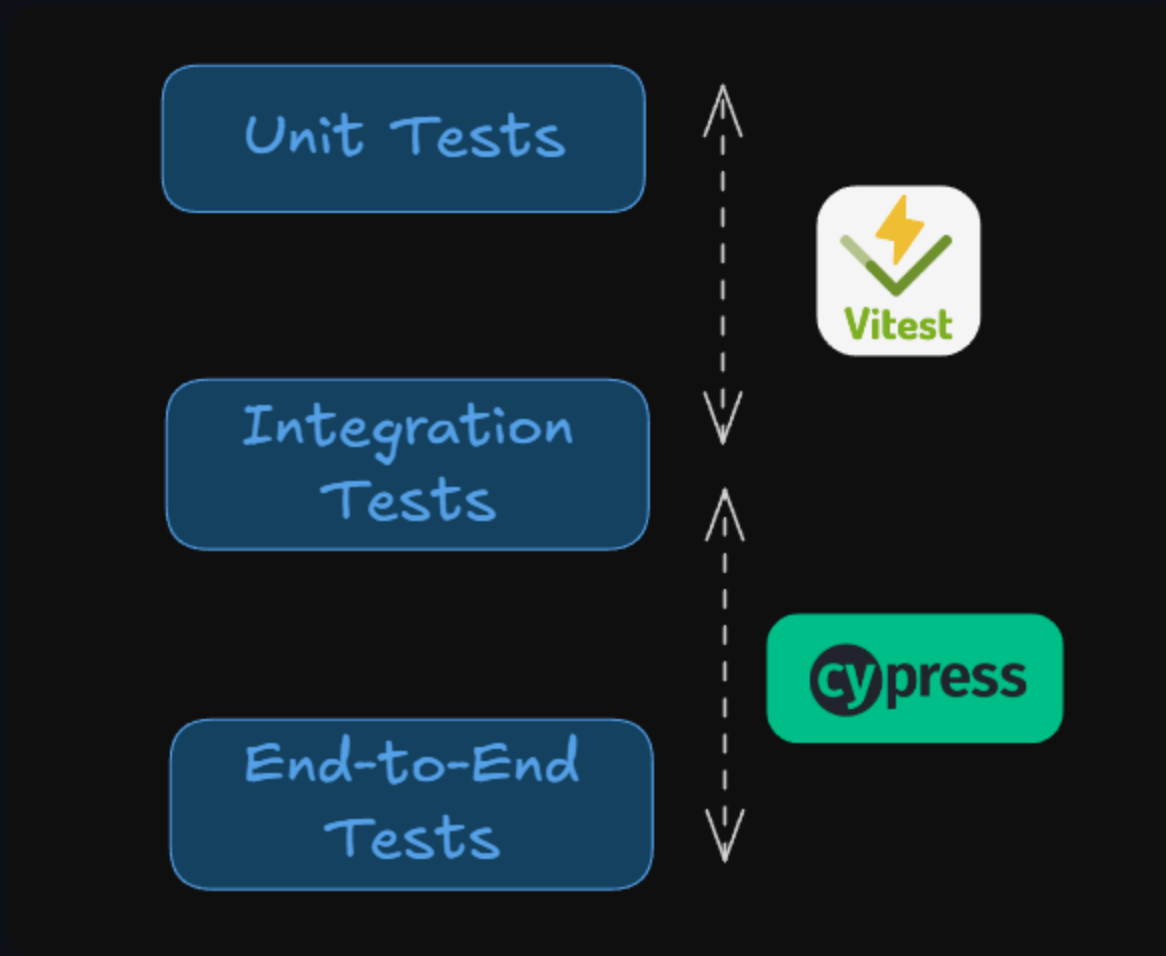
Examples

Test Type	Example	Targeted Scope
Unit Testing	Verifying that <code>add(x, y)</code> returns the sum of <code>x</code> and <code>y</code> .	Individual function/module
Integration Testing	Testing authentication APIs	Combination of modules
End-to-End (E2E) Testing	Simulating a user buying a product.	Full application workflow

Testing frameworks

- State of JS 2025

Testing tools



Storybook?

- Storybook is *overkill* for this project.
- Why?
 - This project has 3 components, so they are easy to track.
 - I am not building a reusable UI library for multiple apps. Therefore, I don't need to maintain a separate Storybook project.

Vitest

Setup

From your `pf-backend` repo

- `pnpm install -D vitest supertest @types/supertest`
- Packages
 - `vitest` is a testing framework similar to Jest.
 - `supertest` is a library for testing HTTP servers.

Setup files

- `package.json` Script ([Link](#))
- `vitest.config.ts` ([Link](#))

Setup Code

- Add at the end of `./src/index.ts`

```
app.listen(PORT, async () => {  
  debug(`Listening on port ${PORT}: http://localhost:${PORT}`);  
});  
  
// Add this line to export the app for testing  
export default app;
```

Test files

```
./src/index.test.ts
```

- Minimal example ([Link](#))
- Full example ([Link](#))

Syntax

- Expect

Cypress

Setting up

- `pnpm init`
- `pnpm install cypress typescript@6 dotenv @tsconfig/node-lts @tsconfig/node-ts @types/node`
- `pnpm approve-builds` (This might take a while to complete.)
- `pnpm exec cypress install` to manually install `Cypress` binary.
- `pnpm exec cypress verify` to verify the installation.

As of July 2026, Typescript 7 does not work with the current version of Cypress (15.18.1). Therefore, we are using Typescript 6.

Files

- `./tsconfig.json` [\(Link\)](#)
- `./.env` from `./.env.example` [\(Link\)](#)
- `./.gitignore` [\(Link\)](#)
- Modify `./package.json` [\(Link\)](#)

cypress in tsconfig.json?

```
{
  "compilerOptions": {
    "types": ["node", "cypress"]
  }
}
```

- The `cypress` type definitions are included in the `tsconfig.json` file to enable TypeScript support for Cypress tests.
 - This allows types of `cy` and `Cypress` to be recognized in the test files.
- The actual definition is located at `node_modules\cypress\types\cypress.d.ts`

Run your first test

- `pnpm run test`

Cypress Browser

- Click `E2E Testing` ➡ `Continue` ➡ `Start E2E Testing in Chrome`
- Click `Create New Spec` ➡ `Name your spec ( min.cy.ts )` ➡ `Run spec`

Minimal example

- `./cypress/e2e/min.cy.ts` [\(Link\)](#)
- Check out [assertion](#) and [actions](#).

Full example

- Enable reading `.env`
 - `./cypress.config.ts` [\(Link\)](#)
- Spec files
 - `./cypress/e2e/backend.cy.ts` [\(Link\)](#)
 - `./cypress/e2e/frontend.cy.ts` [\(Link\)](#)